

Personalized Precision Medicine for Oncology Treatment Optimization

Comprehensive Technical Architecture & Engineering Deliverables Summary | Git Repository Analysis

Repository: https://github.com/nishams63/cancer_analysis	Status: 5 Stages Deployed & Verified	Date: September 2026
Core Frameworks: PyTorch, XGBoost, Transformers, QLoRA, FastAPI, ReportLab, Three.js	GenAI Models: LLaMA-3.2-11B, LLaMA-3.1-70B, BioMistral-7B, Phi-3, BioLinkBERT	Test Suite: 116+ Unit Tests (100% Pass Rate)

1. Executive Summary & Vision

The **Personalized Precision Medicine for Oncology Treatment Optimization** platform is an autonomous, multi-stage clinical AI system designed to improve patient survival, mitigate severe antineoplastic toxicities, predict cancer recurrence, and stress-test clinical AI safety using cutting-edge Generative AI. The repository embodies an enterprise-grade machine learning and deep learning architecture that spans classical tabular predictive modeling (Stage 1), multimodal imaging and longitudinal ctDNA sequencing (Stage 2), negation-aware clinical NLP (Stage 3), fine-tuned Small Language Models (Stage 4), and an advanced GenAI Synthetic Stress-Test Engine with RAG guidance and interactive dashboard hosting (Stage 5).

Each stage was systematically constructed following strict production MLOps standards: zero-leakage data engineering, mathematically rigorous exploratory data analysis (EDA), multi-model benchmarking, independent locked-test evaluation with clinical subgroup audits, containerized FastAPI services, and programmatic Definition-of-Done (DoD) verifications.

Stage	Focus & Discipline	Key Technologies & Models	Primary Deliverable & Impact
Stage 1	Classical Tabular ML Toxicity Prediction	RandomForest, XGBoost, CatBoost, Cox-PH, Optuna, Joblib	Candidate V4 model; predicts severe adverse drug toxicity across 26 cohorts; FastAPI REST service.
Stage 2	Multimodal Deep Learning Progression	Vision CNNs (DenseNet), LSTMs, Transformers, MIL Attention	Fuses histopathology tiles with longitudinal ctDNA time-series for recurrence risk.
Stage 3	Clinical NLP & Concept Extraction	BioLinkBERT, DeBERTa, NegEx Scoping, Rule-based Heuristics	Extracts clinical entities, resolves negation leakage, triages urgent toxicity (Macro F1 0.7557).
Stage 4	SLM Fine-Tuning & Clinical Guidance	BioMistral-7B, Clinical-LLaMA-3, QLoRA, GGUF, llama.cpp	Parameter-efficient fine-tuning; 6-gate clinical safety firewall; 100% offline local CPU inference.
Stage 5	GenAI Synthetic Stress-Test Engine	Monte Carlo Sampler, NVIDIA LLaMA-3.2-11B/3.1-70B, RAG Vector Store	Hybrid statistical + LLM generation; 15 blind spots (BS01–BS15); interactive web dashboard.

2. Detailed Engineering Breakdown by Stage

Stage 1: Classical Machine Learning for Adverse Drug Toxicity Prediction

- **Data Engineering:** Ingested, cleaned, and standardized multi-center oncology patient records into master_patient_dataset.csv. Handled missingness through domain-guided imputation and established schema validations.
- **Exploratory Data Analysis (EDA):** Mapped high-risk biomarker interactions (e.g. baseline creatinine vs platinum clearance), class imbalance across toxicity grades, and generated correlation heatmaps.
- **Model Architecture & Candidate V4:** Benchmarked Random Forest, CatBoost, and XGBoost against baseline heuristics. Regularized the champion **Candidate V4** model to prevent overfitting and serialized via model.joblib.
- **Evaluation & Subgroups:** Evaluated on a strictly locked test set using 95% bootstrap confidence intervals across 26 distinct demographic and clinical cohorts, proving consistent recall across age and gender.
- **Production Integration:** Packaged as a high-throughput FastAPI microservice exposing /health, /predict, and /predict/batch with full Pydantic request validation.

Stage 2: Deep Learning Multimodal Progression & Recurrence Prediction

- **Multimodal Data Pipeline:** Preprocessed gigapixel synthetic pathology tiles and paired them with longitudinal serial ctDNA VAF and tumor marker (CEA, CA-125) time-series sequences.
- **Spatial Deep Learning:** Deployed Convolutional Neural Networks (DenseNet/ResNet) and Multiple Instance Learning (MIL) attention architectures to extract spatial tumor microenvironment features.
- **Temporal Sequence Models:** Built Bidirectional LSTMs and Temporal Transformers capturing velocity and acceleration spikes in circulating tumor DNA before radiological visibility.
- **Multimodal Late Fusion API:** Implemented a late-fusion gating layer combining spatial and temporal embeddings into a single calibrated risk index, wrapped in a dedicated inference API and clinical alert engine.

Stage 3: Clinical NLP & Decision Support Engineering

- **Research Dataset Pipeline:** Generated clinical_nlp_dataset_v1.parquet with automated de-identification (PII scrub), semantic deduplication, and zero-leakage cross-validation splits.
- **20-Section EDA:** Profiled clinical vocabulary, entity distribution, sentence syntax, and created 16 analytical figures in eda_summary.json and exploratory notebooks.
- **Clinical NLP Modeling:** Built negation-aware extractors using BioLinkBERT, DeBERTa, and hybrid clinical heuristics, achieving a Macro F1 score of **0.7557** in acute clinical concept triage.
- **Diagnostic Hardening:** Solved negation scope leakage across contrastive conjunctions ('Denies chest pain, however rapid pulse observed') to ensure accurate symptom triage.

Stage 4: Small Language Model (SLM) Fine-Tuning & Multi-Agent Decision Support

- **4 Engineering Submodules:** Structured across Data Engineer, EDA Engineer, SLM Engineer, and Evaluation Engineer.
- **QLoRA Fine-Tuning:** Fine-tuned open-weights models (BioMistral-7B, Clinical-LLaMA-3, Qwen2.5-1.5B) using 4-bit Quantized Low-Rank Adaptation (QLoRA) on curated oncology instruction pairs.
- **Multi-Axis Evaluation & Safety Firewall:** Subjected models to adversarial clinical prompts, out-of-distribution progress notes, and established a **6-gate safety firewall** (tau* calibration) to block unsafe recommendations.
- **100% Offline CPU Integration:** Exported models to GGUF (Q4_K_M) running locally on CPU via llama.cpp and FastAPI (/summarize), accompanied by a clinician desktop web interface.

Stage 5: GenAI Synthetic Oncology Stress-Test Engine (Master Milestone)

Objective: Create an autonomous adversarial engine that generates statistically valid, biologically grounded, and clinically realistic synthetic oncology patient cases to stress-test and discover edge-case vulnerabilities in Stages 1–4.

Stage 5 was structured across **5 professional engineering roles**:

Role	Scope & Core Technical Responsibilities	Status
Data Engineer	Constructed empirical reference distributions, continuous bounds, biological constraint matrices, and 16 curated evidence chunks (NCCN, CTCAE v5.0, FDA bulletins).	VERIFIED (100%)
EDA / Prompt Engineer	Mined empirical failure modes (FP-001–FP-006), cataloged and ranked 15 blind spots (BS01–BS15), authored prompt library (PROMPT-R01 to R15), and formulated drift immutability rules.	VERIFIED (100%)
Gen AI Engineer	Built two-stage hybrid generation: Monte Carlo StructuredSampler + NVIDIA LLaMA-3.2-11B/3.1-70B NarrativeGenerator, TF-IDF RAG retriever, and counterfactual sensitivity checks.	VERIFIED (100%)
Evaluation Engineer	Constructed 10-step evaluation harness, F01–F20 failure taxonomy, system difficulty scoring, candidate gatekeeper, and 8 comprehensive technical markdown reports.	VERIFIED (100%)
Integration Engineer	Built unified root CLI (run_stage5.py), SQLite WAL result repository (stage5.db), downstream multi-stage adapters, wildcard ranker, and FastAPI dashboard backend.	VERIFIED (100%)

Core Technical Innovations in Stage 5:

- **Hybrid Generation Architecture:** Statistical Monte Carlo sampling ensures 100% biological/pharmacological validity (no hallucinated drug dosages or impossible mutations), while NVIDIA LLaMA-3.2/3.1 creates authentic physician SOAP notes.
- **RAG Grounding & Zero-Hallucination Guardrails:** TF-IDF vector index retrieves approved excerpts from NCCN NSCLC Guidelines v4.2024 and CTCAE v5.0, paired with an automated narrative validator that checks LLM output against the patient's ground truth.
- **Multi-Stage Downstream Stress Testing:** Generated profiles are actively fed through Stages 1, 2, 3, and 4 to uncover blind spots (e.g. Stage 1 tabular model predicting low risk despite metastatic ctDNA spikes; Stage 4 SLM dropping MET bypass resistance).
- **Interactive Localhost Web Platform (Port 8085):** Custom-built interactive UI featuring seed condition selectors, real-time patient synthesis, multi-bar validation scorecards (Statistical 92%, Biological 90%, Clinical 87%), and dynamic scenario distribution donut charts.

3. Interactive Web Dashboards & UI Showcases

The project provides two interactive, user-facing web applications for clinical demonstration and system monitoring:

- 1. Synthetic Patient Generator Dashboard (Stage 5):** Hosted locally on <http://127.0.0.1:8085>. Implements a responsive medical interface with live patient generation, clinical narrative viewer, downstream stress-test harness, scenario history table with CSV export, and analytics charts.
- 2. ONCO.AI 3D Showcase (onco-showcase/):** A modern web application built with React 19, Three.js, and GSAP ScrollTrigger. Delivers a 3D animated walkthrough of the proposed six-stage oncology intelligence workflow with accessible evidence dialogs.

4. Verification, Testing & Codebase Governance

Metric / Audit Criterion	Achieved Value	Verification Mechanism
Automated Unit Test Suites	116+ Tests Passed (100%)	PyTest across Data, EDA, GenAI, Eval & Integration modules
Programmatic Definition-of-Done	100% Passed (Q1–Q10)	Independent verify_done.py audits in all stages
Toxicity Prediction Macro F1	0.7557 (Stage 3 Baselines)	Locked test set with 95% bootstrap confidence intervals
Patient Synthesis Realism	92.4% Statistical, 90.1% Bio	Stage 5 Multi-axis constraint and drift validator
Local Storage & Audit Lineage	100% Deterministic Provenance	SQLite WAL mode (stage5.db) & JSON manifests
Private Offline Deployment	100% Offline CPU Capable	GGUF Q4_K_M + llama.cpp local inference service

5. Viva, Interview & Academic Readiness

The repository contains exhaustive study materials, viva defense guides, and interview transcripts in docs/:

- **Level 1 Viva Preparation Guide:** docs/viva-preparation/Data_Science_Viva_Level_1_Study_Material.pdf covering core ML theory, statistical distributions, regularization, metrics, and cross-validation.
- **Stage 3 & 4 Master Viva Guide:** Detailed HTML and Markdown guides breaking down Clinical NLP tokenization, BioLinkBERT attention mechanisms, LoRA mathematical rank factorization, and tau* calibration.
- **Project Interview Preparation:** docs/PROJECT_INTERVIEW_PREPARATION.md detailing senior engineering answers, trade-off justifications, failure modes, and architectural decisions.

6. Key Operational Commands & Quickstart

Operational Action	Command Line Execution	Output & Target Service
Run Stage 5 Generator	<code>python run_stage5.py --n 20 --seed 42</code>	Generates 20 scenarios, runs stress tests, ranks wildcards into SQLite
Launch Stage 5 Web UI	<code>python run_dashboard.py</code>	Starts FastAPI dashboard server at http://127.0.0.1:8085
Stage 1 Inference API	<code>uvicorn src.app:app --port 8000</code>	Launches Tabular ML toxicity prediction REST endpoints (/predict)
Offline SLM Inference	<code>uvicorn src.app:app --port 8080</code>	Stage 4 GGUF local CPU reasoning service (/summarize)
Run DoD Verification	<code>python "stage 5 Gen-AI/verify_done.py"</code>	Runs automated Definition-of-Done programmatic audits

7. Clinical Impact & Prospective Roadmap

By uniting statistical rigor, deep multimodal embeddings, clinical NLP, localized SLMs, and an adversarial GenAI stress-testing harness, this repository establishes a comprehensive benchmark for AI safety in precision oncology. Immediate roadmap priorities include prospective multi-center cohort validation, direct hospital EHR FHIR integration, and expanding the RAG vector index to encompass international clinical trial protocols (ASCO, ESMO).